

# The AI Powered Developer Platform to Build, Scale, and Deliver Secure Software



Kyungbin Oh

Microsoft Cloud & AI Specialist

# GitHub



**GitHub for Individual**

Free/Pro



**GitHub for Enterprises**

Free/Teams/Enterprise

# GitHub Enterprise



## GitHub Enterprise Server (GHES)

- On-premise deployment
- Except Cloud dependent services



## GitHub Enterprise Cloud (GHEC)

- SaaS Offering
- Powered by AI

# GitHub Enterprise Cloud



## GitHub Enterprise Cloud (classic)

- Personal GitHub accounts id
- SSO/SAML possible
- User Management within GH



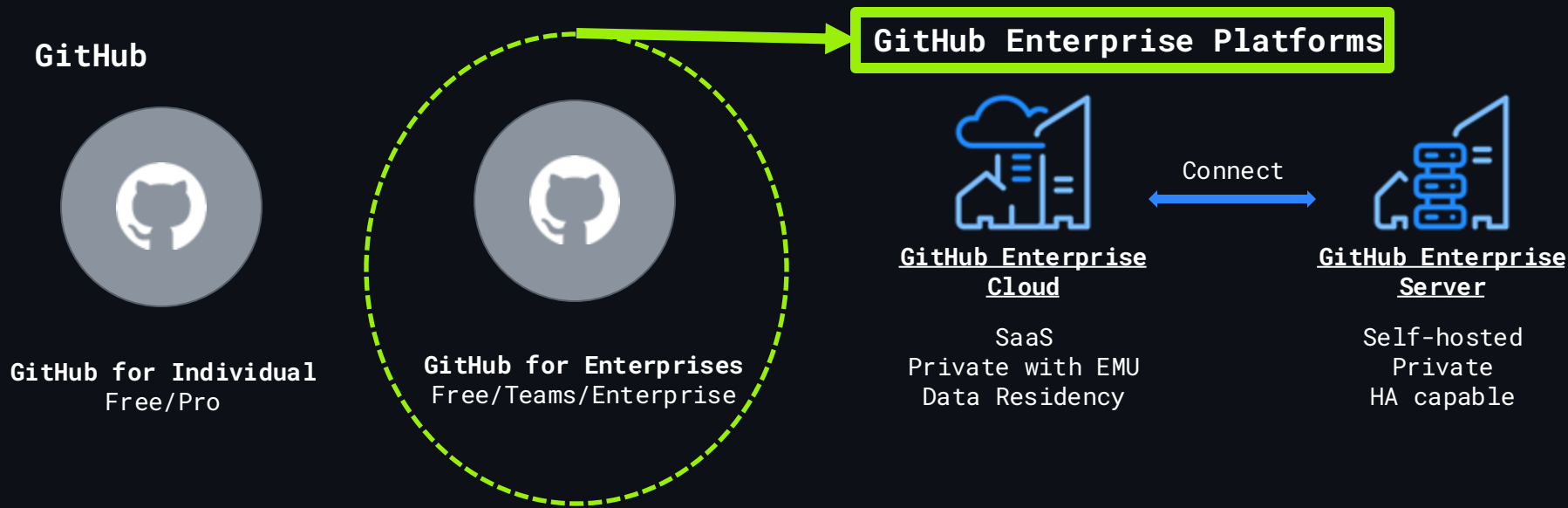
## GitHub Enterprise Cloud with \***EMU**

- Identity Provider (Entra ID/Okta)
- EMU SAML API
- User Management with IdP

*\*Enterprise Managed User (EMU)*

## GitHub Enterprise 란?

GitHub Enterprise는 현대적인 개발의 복잡한 워크플로우를 위해 설계된 Enterprise 급 소프트웨어 개발 플랫폼입니다. 확장 가능한 플랫폼 솔루션으로서, GitHub Enterprise는 조직이 널리 사용되는 IDE, 지속적 통합 도구, 그리고 17,000개 이상의 타사 앱 및 서비스를 지원하여 개발 프로세스 전반에 걸쳐 익숙하고 선호하는 도구들을 원활하게 통합할 수 있도록 합니다.





## GitHub Enterprise Server

- ✓ 고객 네트워크 내 자체 호스팅 방식
- ✓ 다양한 하이퍼바이저 옵션
- ✓ 서브도메인 격리를 통한 XSS 공격 완화
- ✓ 향상된 보안을 위한 아웃바운드 웹 프록시
- ✓ 격리된 사용자 계정
- ✓ GitHub Enterprise Cloud와 대등한 수준의 기능
- ✓ GitHub Connect를 통한 라이선스 동기화 및 기타 기능



## GitHub Enterprise Cloud

- ✓ 클라우드 기반 **SaaS** 서비스
- ✓ 보안 및 정책 관련 기능 제공
- ✓ 새로운 협업자 (Collaborators) 를 위한 빠른 온보딩
- ✓ 줄어든 운영 부담
- ✓ 퍼블릭 리포지토리는 공개됩니다
- ✓ 프라이빗 리포지토리는 접근이 불가능합니다
- ✓ 프라이버시 설정은 Enterprise, Organization, Team, Individual 단계에서 구성



## GitHub Enterprise Cloud + EMUs

- ✓ GHEC (GitHub Enterprise Cloud)는 사용자와 팀 관리를 중앙에서 하게 됩니다 (고객 자체 IdP 활용)
- ✓ **EMU (Enterprise Managed Users)**는 프라이빗 Repository에서 코드를 이동할 수 없습니다
- ✓ EMU의 활동은 GitHub audit log에 기록됩니다

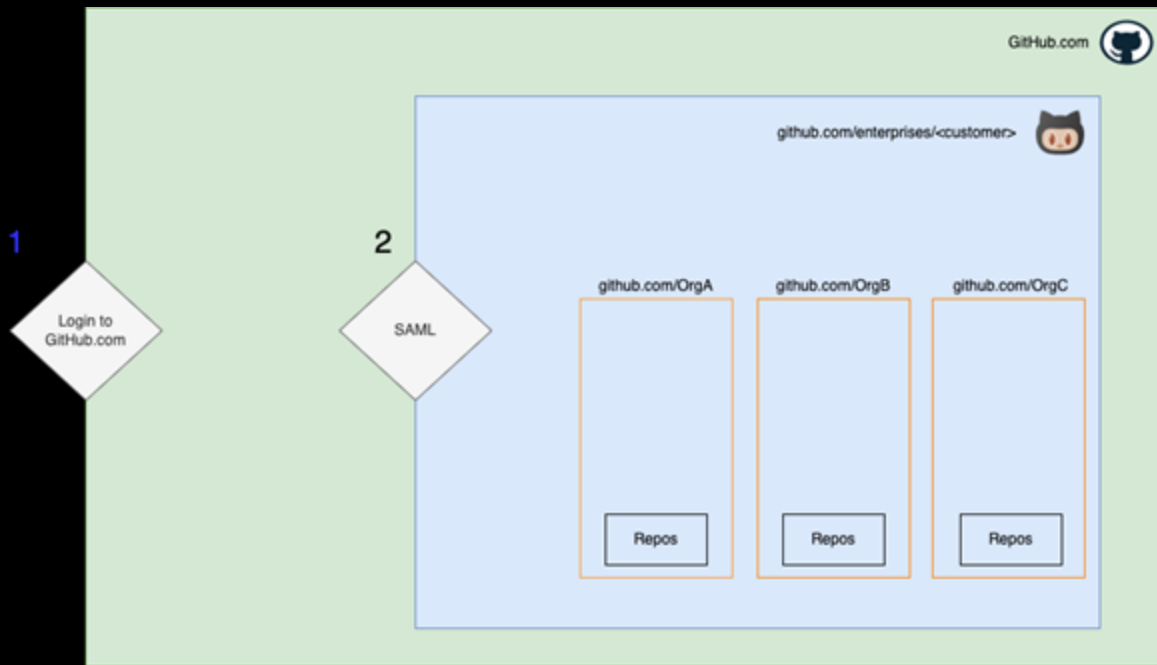


**GitHub Enterprise Cloud with  
personal GitHub accounts (classic)**



# Enterprise SAML

- 현재 및 향후 생성될 모든 조직에 쉽게 적용 가능
- GitHub Enterprise - Enterprise Account Entra ID 앱 사용
- 단 하나의 Entra ID 앱만 관리하면 됨
- 엔터프라이즈 소유자만 구성 가능
- SCIM 미지원 (자동 프로비저닝 기능 사용 불가)
- 사용자가 자신의 계정을 직접 소유

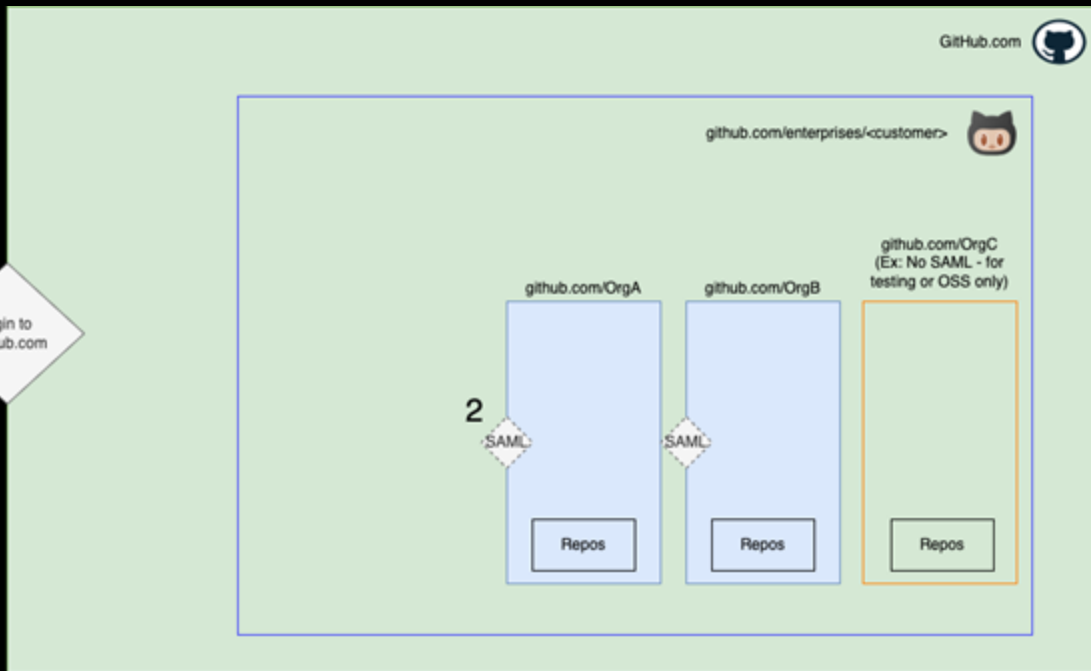




# Organization SAML

- 각 Organization 마다 별도의 SAML Integration 설정 필요
- GitHub Enterprise Cloud - Organization Entra ID 앱 사용
- Organization의 수만큼 Entra ID의 앱을 관리해야 함
- 단, 모든 앱을 동일한 Entra ID 테넌트 또는 여러 테넌트에 연결 가능
- 조직 관리자 (Organization Admin) 만 설정 가능
- SCIM 지원 (자동 프로비저닝 기능 사용 가능)
- 사용자가 자신의 계정을 직접 소유

1





Github.com



 Full access  
GH Resources



@monalisa  
*mona.lisa@octo-inc.com*

SSO Identity Link



GHEC Account

*github.com/enterprises/octo-inc*



IdP

ADFS, AAD, Okta, Ping One,  
Shibboleth



Github.com



Full access  
GH Resources



SSO Identity Link



GHEC Account

github.com/enterprises/octo-inc



1. Login to GitHub  
github.com/login

2. Login to IdP  
per redirect

3. Access  
by role & permissions



IdP

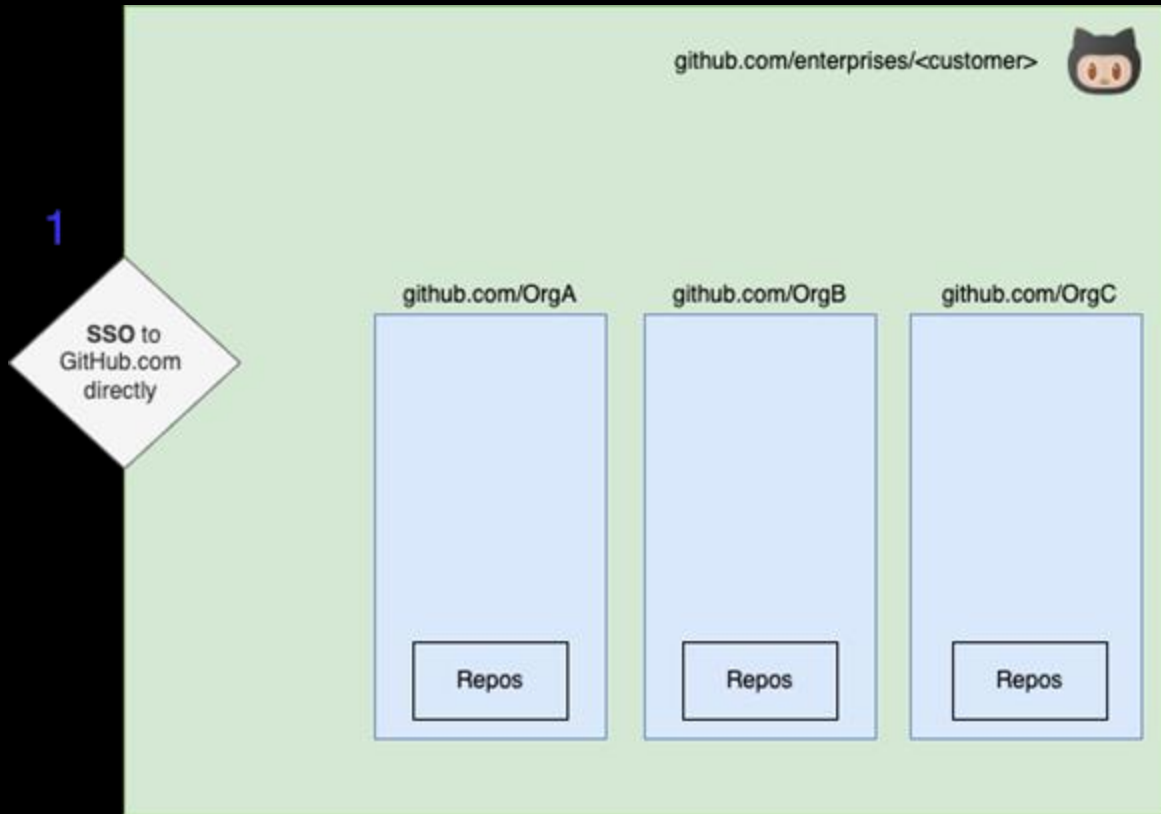
ADFS, AAD, Okta, Ping One,  
Shibboleth

# GitHub Enterprise Cloud with Enterprise Managed Users (EMU)



# Enterprise Managed Users

- Enterprise SAML 방식과 유사
- GitHub Enterprise Managed Users Entra ID 앱 사용
- SAML과 SCIM 모두 필수 구성
- 특수 관리자 계정 (setup user) 만 설정 가능
- 기업이 사용자 계정을 소유 및 관리
- 기존 GHEC 사용자는 새로운 ID를 부여 받음. 모든 코드를 EMU 활성화된 새 엔터프라이즈로 마이그레이션해야 함



# Enterprise Managed Users

중앙화 (Centralized)된 계정 라이프사이클 관리

## GitHub 계정 소유 및 통제

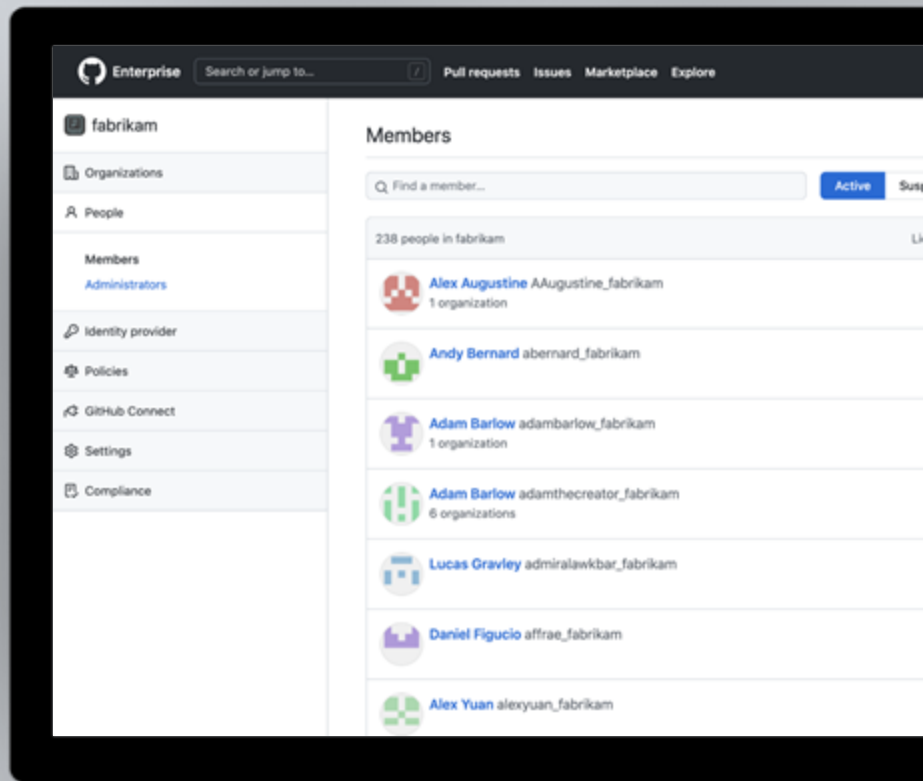
GitHub 관리자가 GHEC 계정을 소유, 생성, 관리 및 감사

## 라이프사이클 자동 관리

ID 공급자를 통해 프로비저닝/프로비저닝 해제 (deprovisioning)  
워크플로우 자동화

## 실수로 인한 IP 유출 방지

회사 소유 콘텐츠의 실수로 인한 공개 노출을 차단하여 IP 유출  
위험 감소





# Enterprise Managed Users

GitHub 계정을 기업이 소유 및 관리

## 업계 표준 프로토콜 활용

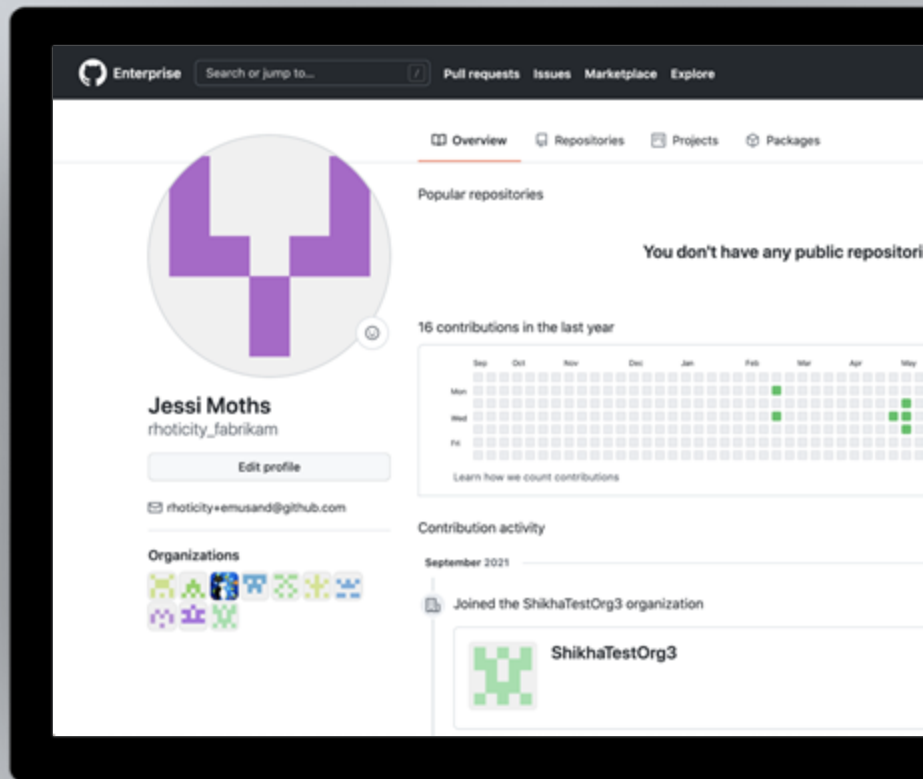
SAML SSO 및 SCIM과 같은 업계 표준 프로토콜을 엔터프라이즈 ID 공급자와 함께 사용

## 프로필 표준화

사용자 이름 및 프로필 정보를 표준화 하여 코드와 상호작용하는 주체를 명확히 파악

## 기업 차원의 약관 동의 (Enterprise Terms & Conditions)

Corporate level에서 이용 약관에 서명



# Enterprise Managed Users

자동화 된 계정 라이프사이클 관리

## IdP를 통한 자동화 된 (de)provisioning

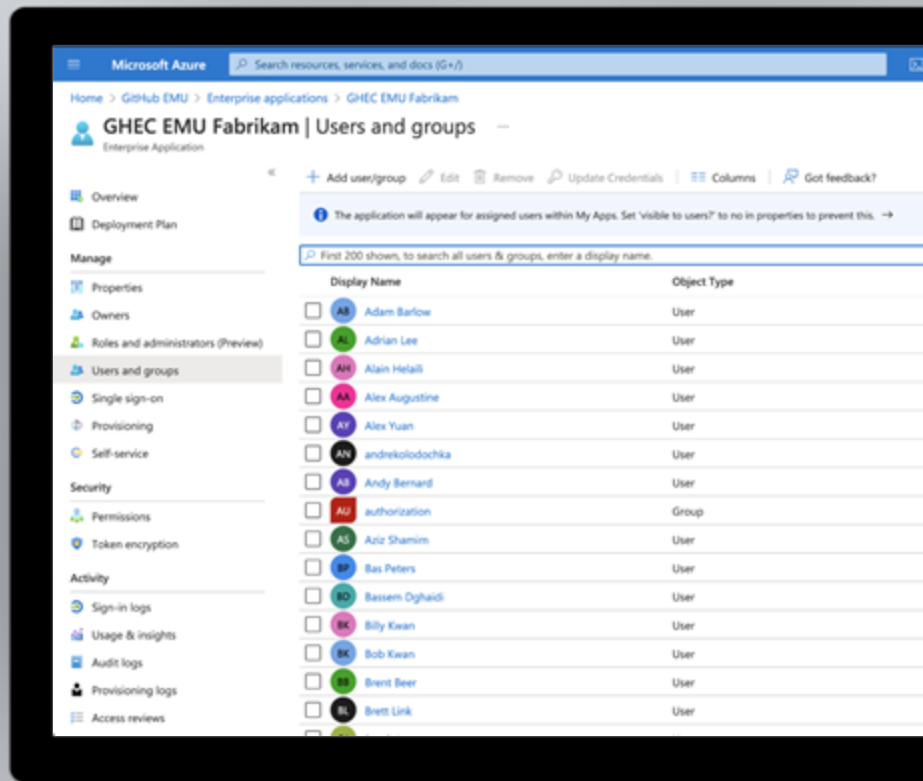
ID 공급자를 사용하여 라이프사이클 관리 자동화

## 통합된 도구 간 워크플로우

다른 소프트웨어 도구의 계정 라이프사이클을 관리하는 것과 동일한 워크플로우 사용

## 직접 할당을 통한 사용자 액세스

직접 할당 방식으로 최종 사용자에게 액세스 제공 - 사용자의 별도 조치 불필요



# Enterprise Managed Users

실수로 인한 IP 유출 감소

## Repository 접근 통제

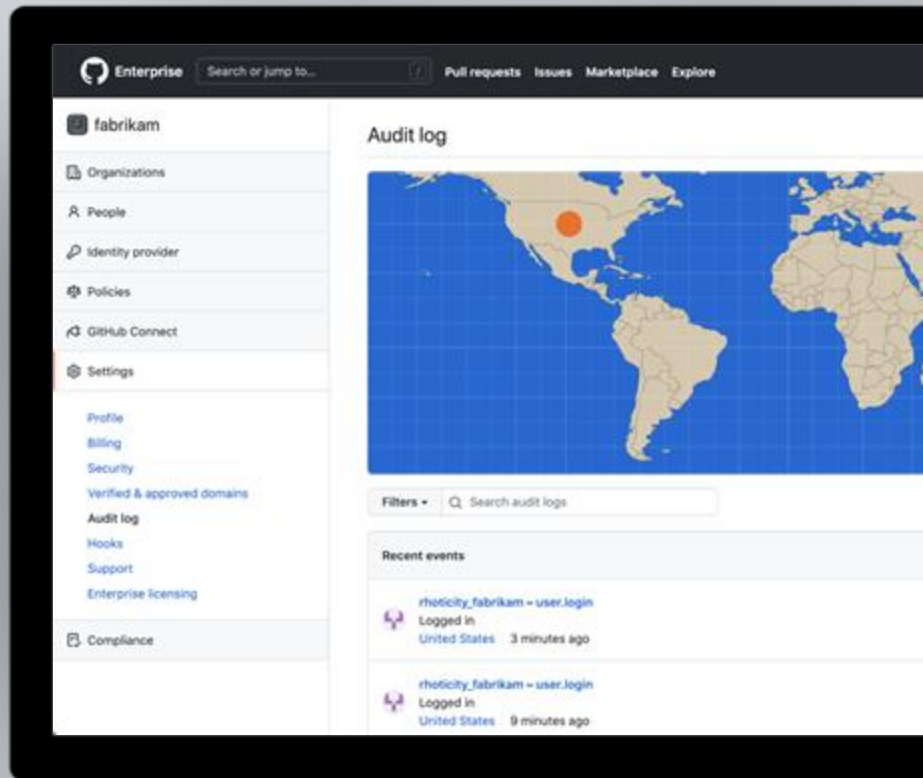
사용자 소유 Repository가 엔터프라이즈 내 다른 사용자에게만 표시되도록 보장

## 감사 로그를 통한 인사이트 확보

User level의 감사 로그 데이터 조회 및 내보내기

## 오픈소스 참여 제한

직원의 계정 정보가 공개 인터넷에 노출되는 행동 방지





Github.com



Read Only  
GH Resources

Enterprise Auth Page



@monalisa\_octo  
*mona.lisa@octo-inc.com*



GHEC+EMU Account

[github.com/enterprises/octo-inc](https://github.com/enterprises/octo-inc)



IdP

AAD, Okta, (Ping)



Read Only  
GH Resources

Enterprise Auth Page



@monalisa\_octo  
*mona.lisa@octo-inc.com*



GHEC+EMU Account  
[github.com/enterprises/octo-inc](https://github.com/enterprises/octo-inc)

1. Navigate to  
enterprise...

[github.com/enterprises/octo-inc](https://github.com/enterprises/octo-inc)



... Login to IdP  
per redirect

2. Access  
by role & permissions



IdP

AAD, Okta, (Ping)



- No public repositories Organization 또는 user 소유 repository를 공개할 수 없음
- No public projects or public GitHub Pages
- No outside collaborators. 모든 사용자 계정은 반드시 IdP를 통해 제공되어야 함
- No public contributions for EMUs (i.e. 엔터프라이즈 계정 외부에서 쓰기 작업 수행 불가)
- No gists 코드 스니펫 공유 기능 사용 불가
- No forking 엔터프라이즈 계정 외부의 Repository 를 포크 (fork)할 수 없음



## Supported options

| <u>Level</u>      | <u>SAML</u> | <u>SCIM</u> | <u>Code migrations needed from existing GitHub org?</u> |
|-------------------|-------------|-------------|---------------------------------------------------------|
| GHEC Enterprise   | ✓ optional  | ✗           | No                                                      |
| GHEC Organization | ✓ optional  | ✓ optional  | No                                                      |
| EMU               | ✓ required  | ✓ required  | <u>YES</u>                                              |



## GHEC



### Control over user account

User in full control over the Account itself



### Multiple IdPs per Organization

ADFS, AAD, Okta, OneLogin, PingOne, Shibboleth



### No control over user activity

User can interact with GitHub (OSS)

## GHEC+EMU



### Control over user account lifecycle

Name, Creation, Suspension



### Single IdP for Enterprise Account

AAD, Okta or Ping



### No public presence or OSS participation

Users are read to GitHub, no public repositories





# GitHub Enterprise Pricing Options

## Free

The basics for individuals  
and organizations

\$0 USD per user/month

- > Unlimited public/private repositories
- > Dependabot security and version updates
- > 2,000 CI/CD minutes/month  
Free for public repositories
- > 500MB of Packages storage  
Free for public repositories
- > Issues & Projects
- > Community support

### FEATURED ADD-ONS

- > GitHub Copilot Access
- > GitHub Codespaces Access

## Team

Advanced collaboration for  
individuals and organizations

\$4 USD per user/month

- ← Everything included in Free, plus...
- > Access to GitHub Codespaces
- > Repository rules
- > Multiple reviewers in pull requests
- > Draft pull requests
- > Code owners
- > Required reviewers
- > Pages and Wikis
- > Environment deployment branches and secrets
- > 3,000 CI/CD minutes/month  
Free for public repositories
- > 2GB of Packages storage  
Free for public repositories
- > Web-based support

### FEATURED ADD-ONS

- > GitHub Secret Protection
- > GitHub Code Security

## RECOMMENDED

## Enterprise

Security, compliance,  
and flexible deployment

Starting at

\$21 USD per user/month

- ← Everything included in Team, plus...
- > Data residency
- > Enterprise Managed Users
- > User provisioning through SCIM
- > Enterprise Account to centrally manage multiple organizations
- > Environment protection rules
- > Repository rules
- > Audit Log API
- > SOC1, SOC2, type 2 reports annually
- > FedRAMP Tailored Authority to Operate (ATO)
- > SAML single sign-on
- > Advanced auditing
- > GitHub Connect
- > 50,000 CI/CD minutes/month  
Free for public repositories
- > 50GB of Packages storage  
Free for public repositories

### EXCLUSIVE ADD-ON

- > Premium support

# GitHub Enterprise Cloud Trial



## GitHub Enterprise Managed Users (EMU) 초기 설정

- GitHub Enterprise Cloud EMU는 GitHub에서 생성하여 조직 내 관리자에게 이메일로 invite 하는 구조 (Self로 설정도 가능함)
- 조직 내 관리자는 EMU Short Code (3-8자리 숫자 또는 문자, 재설정 불가능) 와 Invite를 받을 이메일 주소를 GitHub에 제공해야 함
- 관리자가 사용할 ADMIN 계정은 *Short Code\_admin* 으로 생성됨. 관리자는 이메일에서 Password reset을 진행함.
- (주의) 관리자가 Password 재설정시 반드시 사용하는 브라우저의 Incognito 모드 (InPrivate 모드, 시크릿 모드 등) 으로 설정해야 함.

### Create the setup user

After we create your enterprise, you will receive an email inviting you to choose a password for the setup user, which is used to configure authentication and provisioning. The username is your enterprise's shortcode (chosen by you or randomly generated), suffixed with `_admin`. For example:

`fabrikam_admin`.

Using an **incognito or private browsing window:**

- 1 Set the user's password.
- 2 Enable two-factor authentication (2FA), and save the recovery codes. See [Configuring two-factor authentication](#).

#### Warning

All subsequent login attempts for the setup user account will require a successful 2FA challenge response.

#### Important

If the enterprise account has enabled single sign-on and the setup user hasn't enabled 2FA

# EMU short code란?

- 3~8자의 영문자 및 숫자 조합 (알파벳과 숫자만 사용 가능)
- GitHub 전체에서 고유해야 함 (다른 엔터프라이즈와 중복 불가)
- 모든 사용자 이름에 추가됨 (username\_**short code**)
  - 좋은 예시: kyungbin\_**msft**
  - 권장하지 않는 예시: kyungbin\_**dd38f9a**
- **GitHub에서 Short Code 수정 불가!! 최초 설정 후 변경 불가능하므로 신중하게 선택 필요.**



# GitHub Enterprise Cloud Trial (EMU)

GitHub Enterprise Cloud 서비스는 **30일, 최대 50명이 사용가능한 무료 Trial**이 주어져며, 트라이얼 기간 중 언제든지 정식 Enterprise Cloud로 전환이 가능합니다. Enterprise Cloud로의 전환을 위해서는 Azure의 Subscription ID (구독 ID) 를 등록해야 합니다.

## 1. GitHub Enterprise Cloud Trial (평가판) 링크:

<https://docs.github.com/ko/enterprise-cloud/latest/admin/overview/setting-up-a-trial-of-github-enterprise-cloud>

## 2. GitHub Enterprise with Managed Users (EMU) 옵션 선택

### GitHub Enterprise Cloud 평가판 설치

GitHub Enterprise Cloud의 평가판을 설정하는 방법, 평가판에 포함된 내용 및 평가판이 종료되면 발생하는 일에 대해 알아봅니다.

GitHub Enterprise Cloud는 GitHub에서 협업하는 대기업 또는 팀을 위해 디자인되었습니다. [GitHub Enterprise Cloud 정보](#)(를) 참조하세요.

평가판을 설정하려면 개인 계정에 로그인해야 합니다. 개인 계정이 없는 경우 [GitHub에서 계정 생성하기](#)(를) 참조하세요.

[GitHub Enterprise Cloud의 평가판 설정](#)

30 day free trial

### Unlock GitHub's Premium Features

What you get in your premium free trial:

- ✓ Enterprise: Seamlessly plan, build, and ship great software with an extensible DevOps platform
- ✓ Copilot Business: Improve the developer experience with contextualized, AI-powered coding assistance
- ✓ Advanced Security: Spend less time fixing security vulnerabilities with AI-powered AppSec tools

Recommended for public and private work

#### Enterprise with personal accounts

Recommended for public, open source and private work, that allows members to access your repositories with their [personal GitHub accounts](#).

[Get started](#)

Recommended for internal work

#### Enterprise with managed users

Recommended for private, internal work only, with [company-provided accounts](#) provisioned from your identity provider.

[Get started](#) [For data residency, contact sales](#)

Need more information? Contact your partner or seller. If you don't have one, [get in touch with sales](#).



# GitHub Enterprise Cloud Trial (EMU)

## 3. 내용 작성 후 Create your enterprise 선택 (SHORT CODE 주의!)

30 day free trial

### Try GitHub's Copilot-powered platform

Sign up for your Enterprise Managed Users account

Enterprise name \*

Enterprise URL slug \*

The enterprise account will be available at <https://github.com/enterprises/cloud-magnet>.

**Username shortcode (cannot be edited later) \***

What is a username shortcode? Members of this enterprise will be created with login handles like monalisa\_climaget.

Industry \*      Number of employees \*

Manufacturing      51-1,000

Country \*

Korea, South

Identity Provider \*

☒ Microsoft Entra ID      ☐ PingFederate

☐ Okta      ☐ Custom or other

Select which identity provider you will use to manage your enterprise's user accounts.

---

#### Admin information

This contact will be the first and only admin account responsible to configure the identity provider to provision user accounts and has unique recovery access in the event of an outage.

Your setup admin username will be: monalisa\_climaget

You will set the password for this user in the next step.

Admin name \*      Admin work email \*

4. 신청시 설정한 Admin 이메일 주소로 관리자 초대 메일이 발송됨

5. 관리자 초대 이메일을 통해 관리자 패스워드 재설정

- 반드시 브라우저의 시크릿모드/Incognito 모드/InPrivate모드 활용
- GitHub으로 부터 메일이 2개가 수신되는데, 이중 첫번째로 수신된 패스워드 리셋 링크의 버튼을 클릭



# GitHub Enterprise Managed Users

Microsoft Entra ID를 활용한 SAML SSO 및 SCIM  
Provisioning 설정 가이드



## Prerequisites for GitHub Enterprise SSO Setup

- Microsoft Entra ID tenant
- Entra ID의 아래의 Role assignment가 필요: Application Administrator, Cloud Application Administrator, Application Owner.
- EMU가 활성화 된 GitHub Enterprise + GitHub Enterprise Setup Admin credentials





# SCIM Token 생성하기

The screenshot shows the 'Settings' page for a GitHub Enterprise Cloud account. On the left is a sidebar with navigation links: Overview, Getting started, Organizations, People, Identity provider, Policies, Code Security, Billing & Licensing, Settings, and Compliance. The main content area is titled 'Welcome, [username]' and includes a warning: 'Use this account only to configure single sign-on and SCIM provisioning for the enterprise.' Below this is a vertical list of three steps: 'Generate SCIM token', 'Enable single sign-on', and 'Configure provisioning'. The first step, 'Generate SCIM token', is highlighted with a red rectangular box. It contains the text 'You will need this token to configure provisioning on your identity provider.' and a link 'Learn more about personal access tokens'. A 'Generate SCIM token' button is located at the bottom right of this step. The second step, 'Enable single sign-on', contains the text 'Configure your identity provider so members can authenticate to access organizations.' and a link 'Learn how to configure single sign-on', with an 'Enable single sign-on' button. The third step, 'Configure provisioning', contains the text 'Configure SCIM to automatically provision user accounts to this enterprise after you enable single sign-on.' and a status indicator 'Waiting for SSO configuration' followed by the message 'Unable to detect a tenant connection. Read our guide for configuring provisioning.'

Use this account only to configure single sign-on and SCIM provisioning for the enterprise.

- Generate SCIM token**  
You will need this token to configure provisioning on your identity provider.  
[Learn more about personal access tokens](#)  
Generate SCIM token
- Enable single sign-on**  
Configure your identity provider so members can authenticate to access organizations.  
[Learn how to configure single sign-on](#)  
Enable single sign-on
- Configure provisioning**  
Configure SCIM to automatically provision user accounts to this enterprise after you enable single sign-on.  
Waiting for SSO configuration Unable to detect a tenant connection. [Read our guide for configuring provisioning.](#)



# SCIM Token 생성하기

GitHub Apps

OAuth Apps

Personal access tokens

Fine-grained tokens

Tokens (classic)

New personal access token (classic)

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to [authenticate to the API over Basic Authentication](#).

Getting started / Generate SCIM token

Generate token for SCIM configuration

1. To configure provisioning for your enterprise, you need a personal access token with the "scim:enterprise" scope enabled.

2. Make sure that the token expiration is set to "No expiration". If the token expires, SCIM provisioning may unexpectedly stop working.

[Learn more about creating a personal access token](#)

Note

SCIM provisioning

What's this token for?

Expiration \*  
No expiration ▾ The token will never expire!

Select scopes  
Scopes define the access for personal tokens. [Read more about OAuth scopes](#).

☐ repo

Full control of private repositories

☐ repo:status

Access commit status

☐ repo\_deployment

Access deployment status

☐ public\_repo

Access public repositories

☐ repo:invite

Access repository invitations

☐ security\_events

Read and write security events

☐ workflow

Update GitHub Action workflows

Scope은 아래의 scim:enterprise만 선택

☐ read:discussion

Read team discussions

☐ admin:enterprise

Full control of enterprises

☐ manage\_runners:enterprise

Manage enterprise runners and runner groups

☐ manage\_billing:enterprise

Read and write enterprise billing data

☐ read:enterprise

Read enterprise profile data

☒ scim:enterprise

Provisioning of users and groups via SCIM

☐ admin:gpg\_key

Full control of public user GPG keys

☐ write:gpg\_key

Write public user GPG keys

☐ read:gpg\_key

Read public user GPG keys

☐ admin:ssh\_signing\_key

Full control of public user SSH signing keys

☐ write:ssh\_signing\_key

Write public user SSH signing keys

☐ read:ssh\_signing\_key

Read public user SSH signing keys

Generate token

Cancel



## SCIM Token 생성하기

### Personal access tokens (classic)

Generate new token ▾

Tokens you have generated that can be used to access the [GitHub API](#).



Make sure to copy your personal access token now. You won't be able to see it again!

Continue setup

Copied!



[Blurred token string]

Delete

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to [authenticate to the API over Basic Authentication](#).

Token이 생성되면 별도의 메모에 저장해두기!



# Single Sign-on 구성하기

## Single sign-on configuration

If you centrally manage your users' identities and applications with an identity provider (IdP), you can configure OIDC or SAML to protect your enterprise's resources on GitHub. After you require OIDC or SAML authentication, members of the enterprise must single sign-on with your identity provider (IdP) to access any of the enterprise's organizations.

### OIDC single sign-on

Supported IdPs include Microsoft Entra ID (previously known as Azure AD). [Learn more about OpenID Connect \(OIDC\).](#)

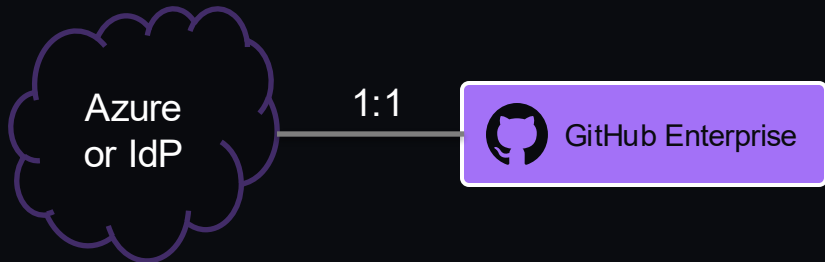
Enable OIDC configuration

### SAML single sign-on

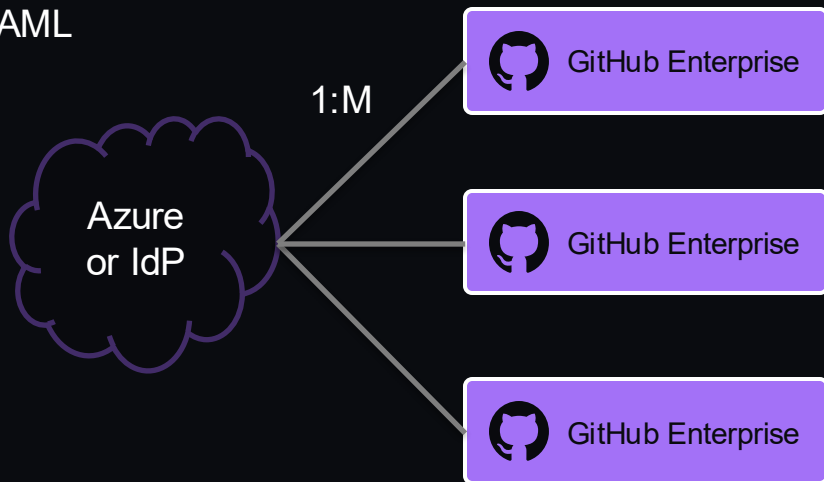
Supported IdPs include Azure, Okta, OneLogin, Ping Identity, or a custom SAML 2.0 provider. [Learn more about SAML single sign-on.](#)

Add SAML configuration

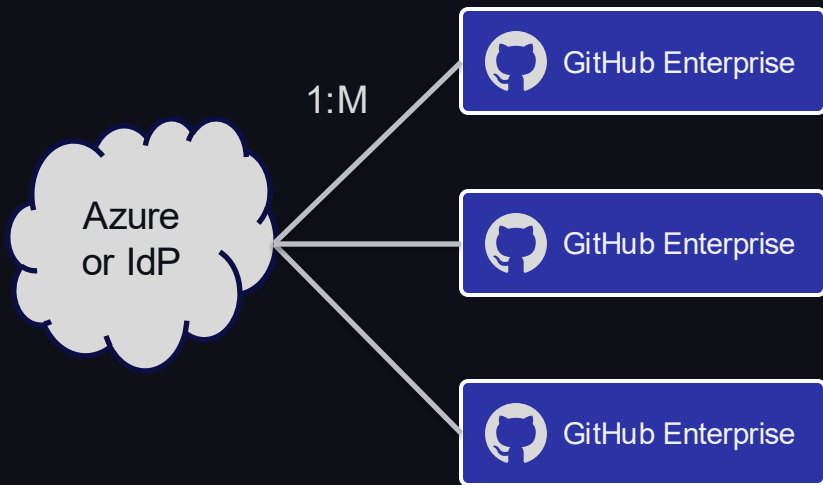
## OIDC



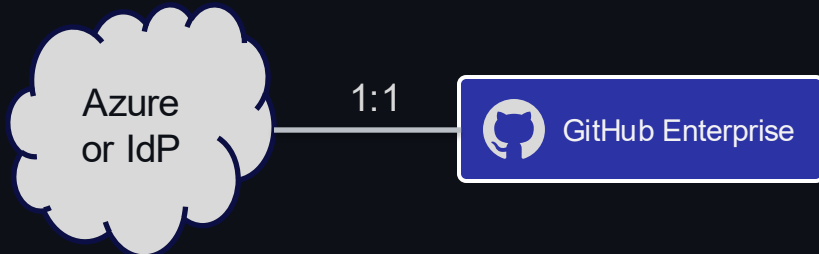
## SAML



## SAML



## OIDC



Microsoft Entra ID를 활용할 경우 OIDC와 SAML 모두 지원합니다.

- OIDC를 활용할 경우 IdP에서 설정한 Conditional Access Policies (CAP)를 지원합니다.
- 하나의 Tenant에 여러 Enterprise를 구성해야 한다면 SAML을 사용해야 합니다 (OIDC는 하나의 Tenant에 하나의 Enterprise 구성만 가능).
- IdP로 Okta를 활용할 경우에는 SAML만 지원합니다.



## Single Sign-on 구성하기

### SAML single sign-on

[← Back to single sign-on configuration](#)

After you require SAML authentication, members of the royalgh enterprise must authenticate with your SAML [about SAML single sign-on](#).

**Sign on URL \***

Members will be forwarded here when signing in to your enterprise

**Issuer \***

Typically a unique URL generated by your SAML identity provider

**Public certificate \***

Paste your x509 certificate here

Your SAML provider is using the  Signature Method and the  Digest Method.

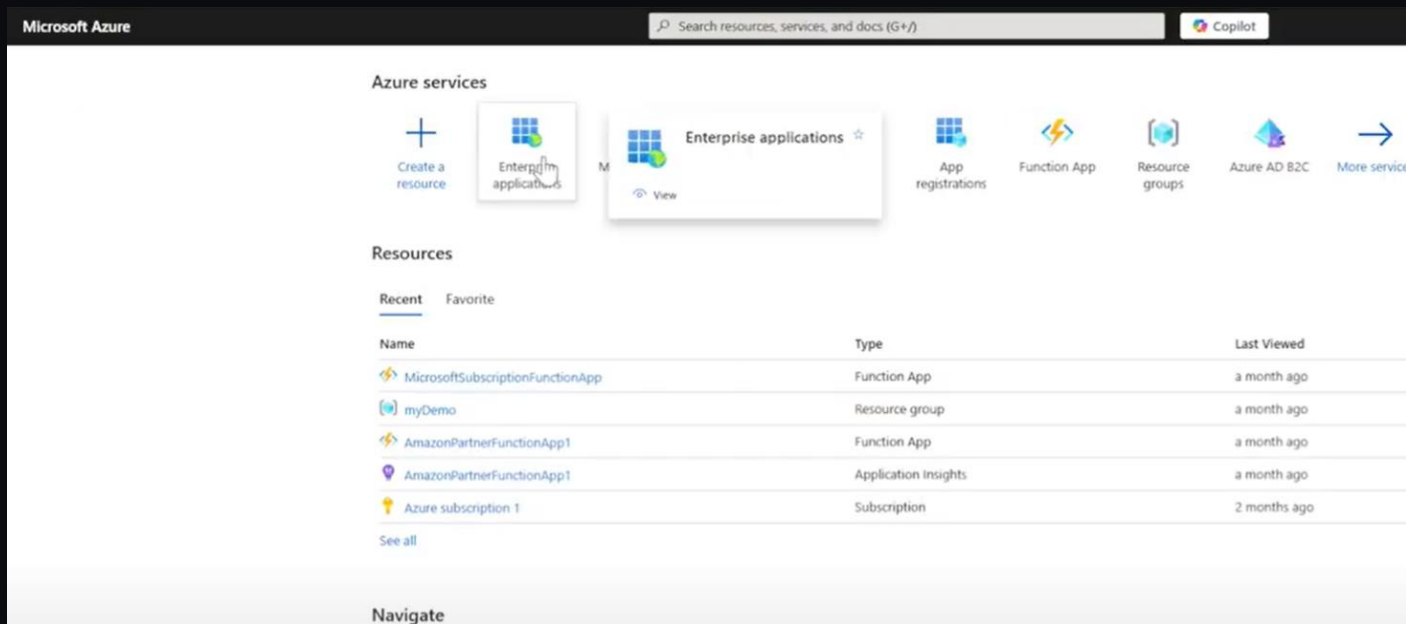
The assertion consumer service URL is <https://github.com/enterprises/royalgh/saml/consume>

⚠ You need to test your SAML configuration before saving.



# Single Sign-on 구성하기 (Azure/Entra ID)

Prerequisite: Microsoft Entra ID의 administrator 권한이 필요  
(application administrator, cloud application administrator, application owner)



Azure Portal - 검색창 - Enterprise Applications



## Single Sign-on 구성하기 (Azure/Entra ID)

Microsoft Azure

Home > Enterprise applications

### Enterprise applications | All applications

Default Directory

+ New application Refresh Download (Export) Preview info Columns Preview features

Overview

Manage

- All applications
- Private Network connectors
- User settings
- App launchers
- Custom authentication extensions

Security

Activity

Troubleshooting + Support

View, filter, and search applications in your organization that are set up to use your Microsoft Entra tenant as their Identity Provider.

The list of applications that are maintained by your organization are in [application registrations](#).

Search by application name or object ID

Application type == Enterprise Applications X Application ID starts with

| Name | App ID | App ID URI | App ID | App ID |
|------|--------|------------|--------|--------|
| ...  | ...    | ...        | ...    | ...    |
| ...  | ...    | ...        | ...    | ...    |
| ...  | ...    | ...        | ...    | ...    |
| ...  | ...    | ...        | ...    | ...    |
| ...  | ...    | ...        | ...    | ...    |
| ...  | ...    | ...        | ...    | ...    |
| ...  | ...    | ...        | ...    | ...    |
| ...  | ...    | ...        | ...    | ...    |
| ...  | ...    | ...        | ...    | ...    |
| ...  | ...    | ...        | ...    | ...    |





# Single Sign-on 구성하기 (Azure/Entra ID)

Microsoft Azure Search resources, services, and docs (G+)

Home > Enterprise applications | All applications >

## Browse Microsoft Entra Gallery

+ Create your own application | Got feedback?

The Microsoft Entra App Gallery is a catalog of thousands of apps that make it easy to deploy and configure single sign-on (SSO) and automated user provisioning. When deploying an app from the App Gallery, you leverage prebuilt templates to connect your users more securely application here. If you are wanting to publish an application you have developed into the Microsoft Entra Gallery for other organizations to discover and use, you can file a request using the process described in [this article](#).

github enterprise manag Single Sign-on : All User Account Management : All Categories : All

Federated SSO Provisioning

Showing 10 of 10 results

**GitHub Enterprise Managed User**  
GitHub, Inc.  
🔗 🔄

**GitHub Enterprise Managed User (OIDC)**  
GitHub, Inc.  
🔗 🔄

**GitHub Enterprise Server**  
GitHub, Inc.  
🔗 🔄

**GitHub Enterprise Server (Legacy)**  
GitHub, Inc.  
🔗 🔄

**GitHub Enterprise Cloud - Organization**  
GitHub, Inc.  
🔗 🔄

**Enterprise Bot**  
Enterprise Bot  
🔗

**SIS Enterprise**  
Sign In Solutions  
🔗

**ArcGIS Enterprise**  
ESRI  
🔗

**Uprise**  
Uprise  
🔗

**Mozy Enterprise**  
EMC Corporation.  
🔗

Microsoft Entra Gallery - 'GitHub Enterprise Managed User' 검색 후 선택



## Single Sign-on 구성하기 (Azure/Entra ID)

Home > Enterprise applications | All applications > Browse Microsoft Entra Gallery >

### GitHub Enterprise Managed User | Overview

Enterprise Application

- Overview
- Deployment Plan
- Diagnose and solve problems
- Manage
  - Properties
  - Owners
  - Roles and administrators
  - Users and groups
  - Single sign-on**
  - Provisioning
  - Self-service
  - Custom security attributes
- > Security
- > Activity
- > Troubleshooting + Support

#### Properties

 Name   
Royal GitHub Enterprise Ma... 

Application ID   
b4ae7cf-d-efad-4fe9-86b6-7... 

Object ID   
c4822c04-6fb9-4285-927f-f... 

#### Getting Started

**1. Assign users and groups**

Provide specific users and groups access to the applications  
[Assign users and groups](#)

**2. Set up single sign on**

Enable users to sign into their appli using their Microsoft Entra credenti  
[Get started](#)

**5. Self service**



# Single Sign-on 구성하기 (Azure/Entra ID)

Microsoft Azure

Search resources, services, and docs (G+/I)

Copilot

Home > Enterprise applications | All applications > Browse Microsoft Entra Gallery > Royal GitHub Enterprise Managed User

## GitHub Enterprise Managed User | Single sign-on

Enterprise application

Overview

Deployment Plan

Diagnose and solve problems

Manage

- Properties
- Owners
- Roles and administrators
- Users and groups
- Single sign-on**
- Provisioning
- Self-service
- Custom security attributes

> Security

> Activity

> Troubleshooting + Support

Single sign-on (SSO) adds security and convenience when users sign on to applications in Microsoft Entra ID by enabling a user in your organization to sign in to every application they use with only one account. Once the user logs into an application, that credential is used for all the other applications they need access to. [Learn more.](#)

### Select a single sign-on method [Help me decide](#)

**Disabled**  
Single sign-on is not enabled. The user won't be able to launch the app from My Apps.

**SAML**  
Rich and secure authentication to applications using the SAML (Security Assertion Markup Language) protocol.

**Linked**  
Link to an application in My Apps and/or Office 365 application launcher.



# Single Sign-on 구성하기 (Azure/Entra ID)

Home > Enterprise applications | All applications > Browse Microsoft Entra Gallery > **GitHub Enterprise Managed User | SAML-based Sign-on** ...

enterprise application

Upload metadata file Change single sign-on mode Test this application Got feedback?

Overview  
Deployment Plan  
Diagnose and solve problems  
Manage  
Properties  
Owners  
Roles and administrators  
Users and groups  
**Single sign-on**  
Provisioning  
Self-service  
Custom security attributes  
Security  
Activity  
Troubleshooting + Support

## Set up Single Sign-On with SAML

An SSO implementation based on federation protocols improves security, reliability, and end user experiences and is easier to implement. Choose SAML single sign-on whenever possible for existing applications that do not use OpenID Connect or OAuth. [Learn more.](#)

Read the [configuration guide](#) for help integrating Royal GitHub Enterprise Managed User.

1

### Basic SAML Configuration

|                                            |          |
|--------------------------------------------|----------|
| Identifier (Entity ID)                     | Required |
| Reply URL (Assertion Consumer Service URL) | Required |
| Sign on URL                                | Optional |
| Relay State (Optional)                     | Optional |
| Logout Url (Optional)                      | Optional |

Edit

2

### Attributes & Claims

⚠ Fill out required fields in Step 1

|                        |                        |
|------------------------|------------------------|
| givenname              | user.givenname         |
| surname                | user.surname           |
| emailaddress           | user.mail              |
| name                   | user.userprincipalname |
| Unique User Identifier | user.userprincipalname |

3

### SAML Certificates

|                           |
|---------------------------|
| Token signing certificate |
|---------------------------|

Edit



# Single Sign-on 구성하기 (Azure/Entra ID)

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

📧 🔔 ⚙️ ? 🗑️

Home > Enterprise applications | All applications > Browse Microsoft Entra Gallery > GitHub Enterprise Managed User | SAML-based Sign-on

Enterprise Application

🔍 ⏪ ⬆️ Upload metadata file ⬇️ Change single sign-on mode 📄 Test this application 🗨️ Got feedback?

Overview

Deployment Plan

Diagnose and solve problems

Manage

- Properties
- Owners
- Roles and administrators
- Users and groups
- Single sign-on**
- Provisioning
- Self-service
- Custom security attributes

Security

Activity

Troubleshooting + Support

Set up Single Sign-On with SAML

An SSO implementation based on federation protocols improves security, reliability, and end user experiences and is easier to implement. Choose SAML single sign-on whenever possible for existing applications that do not use OpenID Connect or OAuth. [Learn more.](#)

Read the [configuration guide](#) for help integrating Royal GitHub Enterprise Managed User.

1

Basic SAML Configuration

Identifier (Entity ID)

Reply URL (Assertion Consumer Service URL)

Sign on URL

Relay State (Optional)

Logout Url (Optional)

Required

Required

Optional

Optional

Optional

Edit

2

Attributes & Claims

⚠️ Fill out required fields in Step 1

givenname

surname

emailaddress

name

Unique User Identifier

user.givenname

user.surname

user.mail

user.userprincipalname

user.userprincipalname

3

SAML Certificates

Token signing certificate

Status

Thumbprint

Expiration

Notification Email

Active

DCF68AE047F204D42EDC2B6362C6B7626850E44D

9/27/2029, 12:57:27 AM

dbarochia@hotmail.com

Edit

Basic SAML Configuration

💾 Save 🗨️ Got feedback?

Identifier (Entity ID) \*

The unique ID that identifies your application to Microsoft Entra ID. This value must be unique across all applications in your Microsoft Entra tenant. The default identifier will be the audience of the SAML response for IDP-initiated SSO.

[Add identifier](#)

Patterns: <https://github.com/enterprise-managed/>

Reply URL (Assertion Consumer Service URL) \*

The reply URL is where the application expects to receive the authentication token. This is also referred to as the "Assertion Consumer Service" (ACS) in SAML.

[Add reply URL](#)

Patterns: <https://github.com/enterprises/<ENTITY>/saml/consume>

Sign on URL (Optional)

Sign on URL is used if you would like to perform service provider-initiated single sign-on. This value is the sign-in page URL for your application. This field is unnecessary if you want to perform identity provider-initiated single sign-on.

Enter a sign on URL ✓

Relay State (Optional)

The Relay State instructs the application where to redirect users after authentication is completed, and the value is typically a URL or URL path that takes users to a specific location within the application.

Enter a relay state

Logout Url (Optional)

This URL is used to send the SAML logout response back to the application.

Enter a logout url ✓

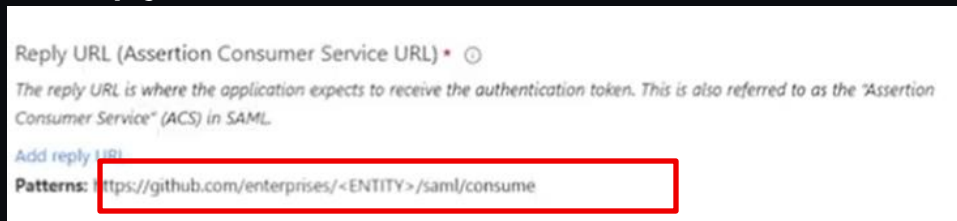


# Single Sign-on 구성하기 (Azure/Entra ID)

## 1. Identifier



## 2. Reply URL



## 3. Sign On URL

<https://github.com/enterprises/<entity>/sso>

**Basic SAML Configuration**

Save | Got feedback?

**Identifier (Entity ID)** \*

The unique ID that identifies your application to Microsoft Entra ID. This value must be unique across all applications in your Microsoft Entra tenant. The default identifier will be the audience of the SAML response for IDP-initiated SSO.

Default

https://github.com/enterprises/<entity>

Add identifier

Patterns: https://github.com/enterprise-managed/\*

**Reply URL (Assertion Consumer Service URL)** \*

The reply URL is where the application expects to receive the authentication tokens. This is also referred to as the "Assertion Consumer Service" (ACS) in SAML.

Index | Default

https://github.com/enterprises/<entity>/saml/consume

Add reply URL

Patterns: https://github.com/enterprises/<ENTITY>/saml/consume

**Sign on URL (Optional)**

Sign on URL is used if you would like to perform service provider-initiated single sign-on. This value is the sign-in page URL for your application. This field is unnecessary if you want to perform identity provider-initiated single sign-on.

https://github.com/enterprises/<entity>/sso

**Relay State (Optional)**

The Relay State instructs the application where to redirect users after authentication is completed, and the value is typically a URL or URL path that takes users to a specific location within the application.



## Single Sign-on 구성하기 (Azure/Entra ID)

### SAML Signing Certificate

Manage the certificate used by Microsoft Entra ID to sign SAML tokens issued to your app

Save + New Certificate ↑ Import Certificate | Got feedback?

| Status | Expiration Date         | Thumbprint                                   |
|--------|-------------------------|----------------------------------------------|
| Active | 11/25/2027, 11:07:28 PM | 02482446F15E8E57774E06ACB28E77F5B4A23A05E... |

Signing Option: Sign SAML assertion

Signing Algorithm: SHA-256

Notification Email Addresses

[Empty text box]

- Make certificate active
- Base64 certificate download
- ↓ PEM certificate download**
- Raw certificate download
- Download federated certificate XML
- Delete Certificate



# Single Sign-on 구성하기 (Azure/Entra ID)

Microsoft Azure

Home > Enterprise applications | All applications > Browse Microsoft Entra Gallery > [Application Name]

## GitHub Enterprise Managed User | SAML-based Sign-on

Enterprise Application

Upload metadata file | Change single sign-on mode | Test this application | Got feedback?

Unique User Identifier: user.userprincipalname

### SAML Certificates

|                             |                                                                                                                       |      |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------|------|
| Token signing certificate   | Active                                                                                                                | Edit |
| Status                      | 02482446F15E8E57774E06ACB28E77E2BAF3A826                                                                              |      |
| Thumbprint                  | 11/25/2027, 11:07:28 PM                                                                                               |      |
| Expiration                  |                                                                                                                       |      |
| Notification Email          |                                                                                                                       |      |
| App Federation Metadata Url | <a href="https://login.microsoftonline.com/751b05fe-aba2-...">https://login.microsoftonline.com/751b05fe-aba2-...</a> |      |
| Certificate (Base64)        | Download                                                                                                              |      |
| Certificate (Raw)           | Download                                                                                                              |      |
| Federation Metadata XML     | Download                                                                                                              |      |

### Verification certificates (optional)

|          |    |      |
|----------|----|------|
| Required | No | Edit |
| Active   | 0  |      |
| Expired  | 0  |      |

### Set up GitHub Enterprise Managed User

You'll need to configure the application to link with Microsoft Entra ID.

|                            |                                                                                                                       |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Login URL                  | <a href="https://login.microsoftonline.com/751b05fe-aba2-...">https://login.microsoftonline.com/751b05fe-aba2-...</a> |
| Microsoft Entra Identifier | <a href="https://sts.windows.net/751b05fe-aba2-4a46-b1...">https://sts.windows.net/751b05fe-aba2-4a46-b1...</a>       |
| Logout URL                 | <a href="https://login.microsoftonline.com/751b05fe-aba2-...">https://login.microsoftonline.com/751b05fe-aba2-...</a> |

Login URL, Identifier, Logout URL을 별도의 메모에 저장해두기!





# Single Sign-on 구성하기

GitHub Enterprise

30 days left on trial. See suggested tasks

- Overview
- Getting started
- Organizations
- People
- Identity provider
  - Single sign-on configuration
  - Groups
- Policies
- Code Security
- Billing & Licensing
- Settings
- Compliance

## SAML single sign-on

← Back to single sign-on configuration

After you require SAML authentication, members of the royalgh enterprise must [sign on](#) before they can access the enterprise.

**Sign on URL \***

Members will be forwarded here when signing in to your enterprise

**Issuer \***

Typically a unique URL generated by your SAML identity provider

**Public certificate \***

Your SAML provider is using the  Signature Method and the  Digest Method.

The assertion consumer service URL is <https://github.com/enterprises/royalgh/saml/consume>

⚠ You need to test your SAML configuration before saving.

GitHub Enterprise Managed User

You'll need to configure the application to link with Microsoft Entra ID.

Login URL

Microsoft Entra Identifier

[Logout URL](#)

- **Sign On URL** = Login URL
- **Issuer** = Microsoft Entra Identifier
- **Public certificate** = PEM certs

# Single Sign-on 구성하기 (Azure/Entra ID - 관리자 setup)

Microsoft Azure

Home > Enterprise applications | All applications > GitHub Enterprise Managed User | Users and groups

Enterprise Application

Overview  
Deployment Plan  
Diagnose and solve problems  
Manage  
Properties  
Owners  
Roles and administrators  
Users and groups  
Single sign-on  
Provisioning  
Self-service  
Custom security attributes  
Security  
Activity  
Troubleshooting + Support

Home > Enterprise applications | All applications > Royal GitHub Enterprise Managed User | Users and groups > Add Assignment

Default Directory

Groups are not available for assignment due to your Active Directory plan level. You can assign individual users to the application.

Users  
1 user selected.  
Select a role \*  
None Selected

Assign users and groups to app-roles for your application here. To create new app-roles, click on the plus icon.

First 200 shown, to search all users & groups, click on the search icon.

Display Name

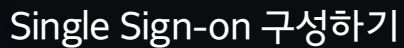
Select a role

Only a single role can be selected

Enter role name to filter items...

Billing Manager  
Enterprise Owner  
Restricted User  
User

Enterprise Applications - GitHub Enterprise Managed User - Users and Groups - Add user/group - 관리자 할당 진행 (Role: Enterprise Owner)



[← Back to single sign-on configuration](#)

Sign on URL \*

<https://login.microsoftonline.com/751b05fe-aba2-4a46-b1e0-a3bff>

Members will be forwarded here when signing in to your enterprise

Issuer 

<https://sts.windows.net/751b05fe-aba2-4a46-b1e0-a3bf80b94e7e/>

Typically a unique URL generated by your SAML identity provider

Public certificate <sup>a</sup>

-----BEGIN CERTIFICATE-----  
MIIC8DCCAdigAwIBAgIQZqc3Yw1Rb6ple8+Iixop4zANBgkqhkiG  
9w0BAQsFADAOMTIwMAYDVQQDDA  
EylNaWNyb3NvZnQxcmUgRmVtZXJhdGVkfENTTyBDXJO  
aWZyP2FOZTAeFwoYNDEzMjUxNmM3  
MijlaFw0YnZlExMjUxNmM3MDQxMjAwBgcqNVBAMTKU1pY3lv  
c29mdCRRenVuZWScZWRlcmljaHQwOg

Your SAML provider is using the **RSA-SHA256** Signature Method and the **SHA256** Digest Method.

The assertion consumer service URL is <https://github.com/enterprises/royalgh/saml/consume>

### Test SAML configuration

✔ Passed: Successfully authenticated your SAML SSO identity.



## Single Sign-on 구성하기

### Single sign-on recovery codes

Put these in a safe spot. If your identity provider is unavailable and you don't have a recovery code you will lose access. A recovery code grants access for 24 hours and can only be used once.

#### Recovery codes

Treat your recovery codes with the same level of attention as you would your password! We recommend saving them with a password manager such as [1Password](#), [Authy](#), or [Keeper](#).



- IdP에 문제가 있을 경우 Recovery Codes를 사용해서 GitHub 엔터프라이즈 계정에 로그인할 수 있습니다.
- 사전에 Recovery Codes를 다운로드하고 안전하게 보관해야 합니다.
- 엔터프라이즈 관리자 (Enterprise Owner) 만 사용 가능합니다 (EMU의 경우 Setup User).
- 각각의 Recovery Codes는 1회만 사용 가능하며, 순서대로 사용해야 합니다. 사용 후 24시간 동안만 접근을 허용하고 있습니다.
- [Troubleshooting identity and access management for your enterprise - GitHub Enterprise Cloud Docs](#)

Enable SAML authentication



# SCIM Provisioning 구성하기 (Azure/Entra ID)

Microsoft Azure

Home > Enterprise applications | All applications

## GitHub Enterprise Managed User | Provisioning

enterprise application

Overview

Deployment Plan

Diagnose and solve problems

Manage

- Properties
- Owners
- Roles and administrators
- Users and groups
- Single sign-on
- Provisioning**
- Self-service
- Custom security attributes

Security

Activity

Troubleshooting + Support

The application will appear for assigned users within My Apps. Set 'visible' to 'true' to make the application visible to users.

Microsoft Azure

Home > Enterprise applications | All applications

## Provisioning

Save Discard

Provisioning Mode

Automatic

Use Microsoft Entra to manage the creation and synchronization of user accounts in Royal GitHub Enterprise Managed User based on user and group assignment.

Admin Credentials

Admin Credentials

Microsoft Entra needs the following information to connect to Royal GitHub Enterprise Managed User's API and synchronize user data.

Tenant URL \*

`https://api.github.com/scim/v2/enterprises/{enterprise}`

Secret Token

`SCIM TOKEN`

Test Connection

Settings



## SCIM Provisioning 구성하기 (Azure/Entra ID)

**Provisioning** ...

Save X Discard

Endpoint URL

Secret Token

Test Connection

^ Mappings

Mappings

Mappings allow you to define how data should flow between Microsoft Entra ID and GitHubEnterpriseCloud.

| Name                                | Enabled |
|-------------------------------------|---------|
| Provision Microsoft Entra ID Groups | Yes     |
| Provision Microsoft Entra ID Users  | Yes     |

☐ Restore default mappings

^ Settings

☒ Send an email notification when a failure occurs

Notification Email \*

☐ Prevent accidental deletion

Scope

Sync only assigned users and groups

Provisioning Status

On Off

- Mapping: Entra ID Groups/Users - YES
- Scope: Sync only assigned users and groups
- Provisioning Status: ON

# GitHub Copilot



## GitHub Copilot Business 제공

- GitHub Copilot Standalone은 Trial로 제공되지 않으며 GitHub Enterprise Cloud Trial 시 **GitHub Copilot Business 제공 (30일, 50명)**
- Enterprise Account에서 Azure Subscription을 연동하여 활용 ([Connecting an Azure subscription - GitHub Enterprise Cloud Docs](#))
- 혹은 신용카드 정보를 등록하여 활용 가능. Trial은 자동으로 deactivate되나 혹시 모를 과금을 방지하기 위하여 30일차에 GitHub Enterprise Cloud 삭제 권고
- Enterprise Account 관리 페이지에서 Organization을 생성하고 GitHub Copilot Business 라이선스를 제공할 사용자들을 추가 (EMU로 구성된 경우 연동한 IdP의 사용자들은 Provisioning 되어 있는 상태)





# GitHub Copilot Business vs GitHub Copilot Enterprise

## Business

Accelerate workflows with GitHub Copilot.

**\$19**<sup>USD</sup>

per user / month

Get started

Contact sales

### ^ What's included:

- ✓ Unlimited agent mode and chats with GPT-5 mini<sup>1</sup>
- ✓ Unlimited code completions
- ✓ Access to code review, Claude Sonnet 3.5/3.7/4, Gemini 2.5 Pro, and more
- ✓ 300 premium requests to use latest models per user, with the option to buy more<sup>2</sup>
- ✓ User management and usage metrics
- ✓ IP indemnity and data privacy
- ✓ Coding agent

## Enterprise

Best value

Scale with AI agents and comprehensive model access.

**\$39**<sup>USD</sup>

per user / month

Get started

Contact sales

### ^ Everything in Business and:

- ✓ Access to all models, including Claude Opus 4.1, o3, and more
- ✓ 3.33x more premium requests than Business to use the latest models, with the option to buy more<sup>2</sup>
- ◆ Access to [GitHub Spark](#)



**Thank you**